

Jesse Hurtado

Pushing productivity forward through data driven solutions

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Spoken Languages: English (Native Fluency), Spanish (Native Fluency)

EDUCATION

University of California, Los Angeles (UCLA) | Los Angeles, California

Bachelor of Science | Computational Neuroscience, Graduated Spring 2023

San Diego State University (SDSU) | San Diego, California

Master of Science | Big Data Analytics, Expected Graduation Spring 2026

GPA: 4.0 (4.0 Scale)

PROJECTS

Alvarado Creek: Time Series Analysis Tracks Water Quality Dynamics

- Developed a MATLAB-based time series analysis pipeline to clean and correct real-time high-frequency fluorescence spectroscopy data, removing sensor artifacts and anomalies to ensure accurate water quality monitoring in urban headwater streams. <https://wqdatalive.com/public/v3/1869?dashboardId=309>

Medical Machine Learning: Predicting and Mapping Diabetes Cases in San Diego County

- Developed a multivariate linear regression model to predict future diabetes cases in San Diego County and built a Tableau-powered analytics dashboard with a dynamic demographic heat map for data visualization.
- <https://sites.google.com/view/mmlbda/home?authuser=0>

Neural Place Cell Imaging - Constructing 2D Histograms Reveal Spatial Neural Activity

- Developed MATLAB analysis functions to process, visualize, and statistically analyze hippocampal CA1 neuron activity, investigating the role of place cells in spatial navigation during experimental trials.

Point Source Approximation - Simulating Neural Activity with Non-Linear Differential Equations

- Developed a MATLAB simulation using nonlinear differential equations to model 2 neurons firing in a homogeneous extracellular medium, applied a point source approximation to estimate the current distribution at a hypothetical distance.

EKG Power Analysis

- Developed a Python-based system to operate an Arduino and micro-EKG simultaneously during bench press workouts, analyzing voltage changes during the concentric phase of movement.
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WORK EXPERIENCE

Graduate Research Assistant II – *SDSU Water Innovation and Reuse Lab*, September 2024 - Present

Supervisor: Dr. **Natalie Mladenov** Ph.D, P.E., nmladenov@sdsu.edu

Reference: Dr. **Kenisha Shipley** , Ph.D, kshipley@sdsu.edu

Conduct data analysis and visualization of creek water quality data using MATLAB and Python to identify trends and anomalies while maintaining and calibrating real-time sensor platforms for accurate monitoring. Assist in manuscript writing and contribute to an ongoing research paper on water quality assessment. Working toward developing an autonomous correction system to improve the accuracy of real-time water quality monitoring by detecting and correcting sensor anomalies.

Undergraduate Research Assistant I – *UCLA Center for Neurophysics*, January 2023 - July 2023

- Studied spatial mapping activity of hippocampal CA1 neurons, focusing on data processing and signal analysis. Developed software libraries for experimental trials and worked to automate data processing algorithms to improve efficiency and reproducibility.

Applied Math Tutor – *UCLA Curtis Center for Mathematics / Barack Obama Prep Academy*, August 2022 - December 2023

- Tutored 7th and 8th grade mathematics in both English and Spanish to students in underserved communities, focusing on building critical thinking skills and strengthening problem-solving abilities through an applied math curriculum, emphasizing basic algebraic and statistical literacy.

Data Analyst Intern – *Kaiser Permanente, Release of Medical Information Department*, June 2022 - August 2022

- Collected, processed, and analyzed quantitative and qualitative data using Excel and SQL to assess efficiency in medical records request intake and fulfillment workflows. Tracked key metrics such as request type, processing time, department origin, and request volume (hourly, daily, weekly) to identify bottlenecks and recommend workflow optimizations for improved departmental efficiency.

SKILLS AND TECHNICAL TOOLS

MATLAB - Advanced proficiency in **signal processing**, **quantitative analysis**, and **data visualization** for engineering and research applications:

- **Signal Processing:** Analyze, filter, and transform signals using MATLAB's **Signal Processing Toolbox**, implementing FFT, wavelet transforms, and noise reduction techniques.
- **Exploratory & Descriptive Statistical Analysis:** Perform in-depth statistical analysis, including **hypothesis testing**, **regression modeling**, and **data clustering** to extract meaningful insights.
- **Data Visualization:** Develop clear and insightful visualizations using MATLAB's plotting tools to represent complex datasets and experimental results.

PYTHON - Advanced proficiency in Python for data analysis, statistical modeling, and machine learning using industry-standard libraries:

- **Data Analysis & Visualization:** Extensive experience with **pandas**, **numpy**, and **matplotlib/seaborn** for data wrangling, statistical analysis, and creating insightful visualizations.
- **Predictive Modeling & Machine Learning:** Experience developing and optimizing **classification** and **regression models** using **scikit-learn** and **PyTorch**, leveraging techniques such as feature engineering, hyperparameter tuning, and cross-validation
- **Data Structures & Algorithms:** Strong understanding of **hash tables**, **trees**, **graphs**, **linked lists**, and **search/sorting algorithms** to optimize data processing and analysis workflows.